

J. Stephen Huang

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Education

University of Waterloo, BSc in Computer Engineering – Waterloo, ON

Sep. 2024 – Apr. 2029

- GPA: 3.9/4.0

Experience

Software Engineer Intern, [Fellow](#) – Montreal, QC

May 2026 – Aug. 2026

- Strengthened workspace security by implementing admin controls for MCP tools and integrations using Python, Django, and GraphQL, allowing organizations to restrict users to approved tools and reducing the risk of malicious data export through user integrations.
- Integrated Procore, a construction management platform, into Fellow.ai, giving construction teams a single entry point to access Procore assets, meeting notes, and documents directly within their meeting workflow.
- Resolved 10-15 customer-reported bugs and product requests as they arose, improving reliability and satisfaction for 5k+ customers and enabling the sales team to close deals and secure new clients.
- Implemented redaction of PII and sensitive data across the Django and React monorepo, reducing customer data exposure in internal tools and workflows.

Software Engineer Intern, [Parametric Research Labs](#) – Toronto, ON

Sep. 2025 – Dec. 2025

- Developed 2 AI voice agents capable of fully handling 30 inbound and outbound calls in parallel using AWS, LiveKit, and NestJS, reducing routine call handling for high-volume clinics by 70% and lowering labor costs by 50%.
- Built a conversational SMS agent leveraging OpenAI's API with function calling to deliver dynamic reminders, follow-ups, call summaries, and rescheduling options to patients, greatly enhancing user experience before and after calls.
- Engineered an AWS pipeline that validates uploaded call list CSVs and automatically schedules outbound calls, saving the clinic over 1,000 manual call launches per week.
- Built a local and automated test suite for voice and SMS agents, accelerating pre-deployment validation, reducing the risk of production downtime, and saving approximately \$1 in testing costs per deployment.

Software Engineer, [Garment System](#) – Montreal, QC

Feb. 2025 – Present

- Implemented a full-stack platform using Next.js, Redis, and Prisma to streamline clothing batch requests from 100+ clients, enabling requests to be fulfilled 50% faster through an intuitive interface and clear data visualization.
- Engineered a fully embedded Stripe checkout experience that allows customers to purchase clothing worldwide, accurately handling tax, shipping, and order creation, and successfully processing 50+ payments on launch day.
- Resolved 40+ high- and medium-priority bugs, ensuring uninterrupted operations for the manufacturing team.

Mechanical Engineer Intern, [Sheartak Tools](#) – Waterloo, ON

Jan. 2025 – Apr. 2025

- Designed 30+ precise and detailed 3D models and 2D technical drawings in SolidWorks for new product lines, custom tools, and existing inventory, providing the manufacturing team with production-ready specifications.
- Authored clear and accurate installation manuals for 20+ products, enabling customers to complete installations confidently and independently, significantly reducing technical support inquiries.

Robotics Software Engineer, [WATonomous](#) – Waterloo, ON

Sept. 2024 – Present

- Built C++ world modeling modules that ingest 3D object detections from perception into context-rich HD map objects.
- Implemented ROS2 publishers and services to expose world model state such as traffic light states, rules, and lane context to downstream decision-making and path-planning.
- Designed and implemented a C++ behaviour tree that translates world model data into driving commands for the control stack, enabling the vehicle to navigate safely toward requested goal points while obeying traffic rules.
- Validated the end-to-end world modeling, behaviour planning, and control pipeline in CARLA with the planning team, identifying integration issues before vehicle testing and saving an estimated 20+ hours of test track time.
- Created a Lanelet2-compatible OSM file of the test track using Autoware tools, enabling global route planning.

Projects

2026 World Cup Prediction Pipeline

- Engineered and selected match features from historical international soccer data to train a Random Forest classifier using scikit-learn and Python, predicting soccer match outcomes with a Brier score of 0.082.
- Ran 20,000 Monte Carlo simulations of the 2026 World Cup using the trained model, correctly predicting Spain as champion and Argentina as runner-up, helping me win a 25-participant prediction pool with a \$400 prize.

Interpreter in C++

- Built an interpreter in C++17, implementing lexical analysis, parsing, and execution to translate Lox syntax source code into program behaviour with my own twists and optimizations.

GenAI YouTube Automation Pipeline

- Created a Python pipeline with open-source libraries to automate end-to-end narration video creation, eliminating the need to manually source content, record narration, edit, and upload, saving 10 hours per video at zero cost.
- Built a LangGraph workflow with checkpointing that preserves progress between stages, allowing interrupted runs to resume without repeating completed work and supporting human-in-the-loop validation of intermediate results.

AI Transcription Platform

- Led the development of a Next.js web application that provides 99% accurate AI transcriptions for content creators.
- Integrated robust Firebase rules for user authentication and data storage, and Stripe for secure payment processing.

Skills

Languages: C++, Python, C, Bash, SQL, TypeScript, JavaScript

Robotics: ROS 2, Lanelet2, BehaviorTree.CPP, CARLA, Foxglove, OpenStreetMap

ML & AI: OpenCV, YOLO, scikit-learn, LangChain, LangGraph

Web: Django, React, Next.js, NestJS, Prisma

Platforms: Linux, Docker, AWS, GitHub, GitHub Actions, Vercel

Services & Data: OpenAI, LiveKit, Supabase, Firebase, Twilio, Stripe, Redis, Cekura

Hardware: SolidWorks, STM32CubeIDE, ESP-IDF